CC7b Allotropes of Carbon

Learning Outcomes:

CC1.34 Recall some allotropes of carbon.

CC1.34 Describe the basic differences between covalent, simple molecules and giant covalent structures.

CC1.35 Describe the structures of diamond, graphite, fullerenes and graphene.

CC1.36 Describe the properties of diamond, graphite, fullerenes and graphene.

CC1.36 Explain the properties and uses of diamond and graphite in terms of their structure and bonding.

CC1.37 Explain the properties of fullerenes and graphene in terms of their structure and bonding.

How many compounds and materials can you think of that contain Carbon.

Molecules

Molecules are groups of …**atoms**… joined by …**covalent**….. bonds. They can be …**compounds**.. made up of more than one type of atom or …**elements**…made up of the same type atom.

KEYWORDS:

Compounds

Atoms

Elements

Allotropes

Allotropes are different **structural** forms of a particular **element**.

The **structure** and **bonding** in different allotropes influences their **properties** and **uses**.

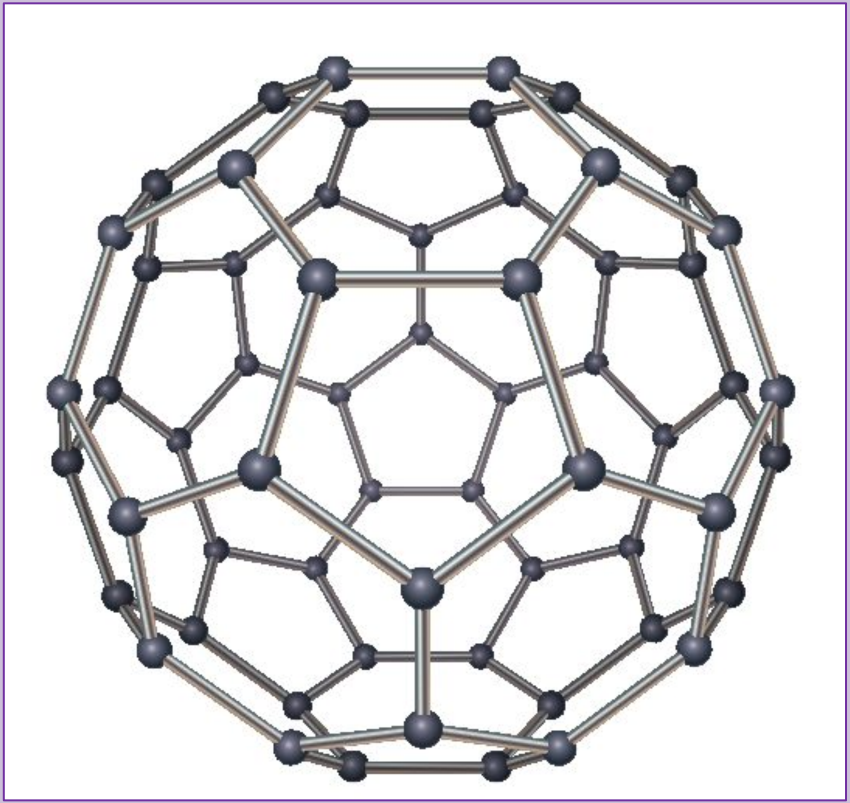
	Simple covalent structure or Giant covalent structure	Structure	Bonding	Properties and Uses
Diamond				
Graphite				
Graphene				
Fullerenes				

FULLERENES

In **simple molecules** called fullerenes, carbon atoms are **covalently** bonded to three other carbon atoms.

They are often **tubular** molecules or **spherical**.

Fullerenes have **weak intermolecular forces** between the molecules so have **low melting points**.
These weak forces also make them **soft and slippery**.
However the molecules themselves are very **strong** due to their covalent bonding.



C_{60} or Bucky Ball
An example of spherical fullerene which has 60 carbon atoms that form a ball.

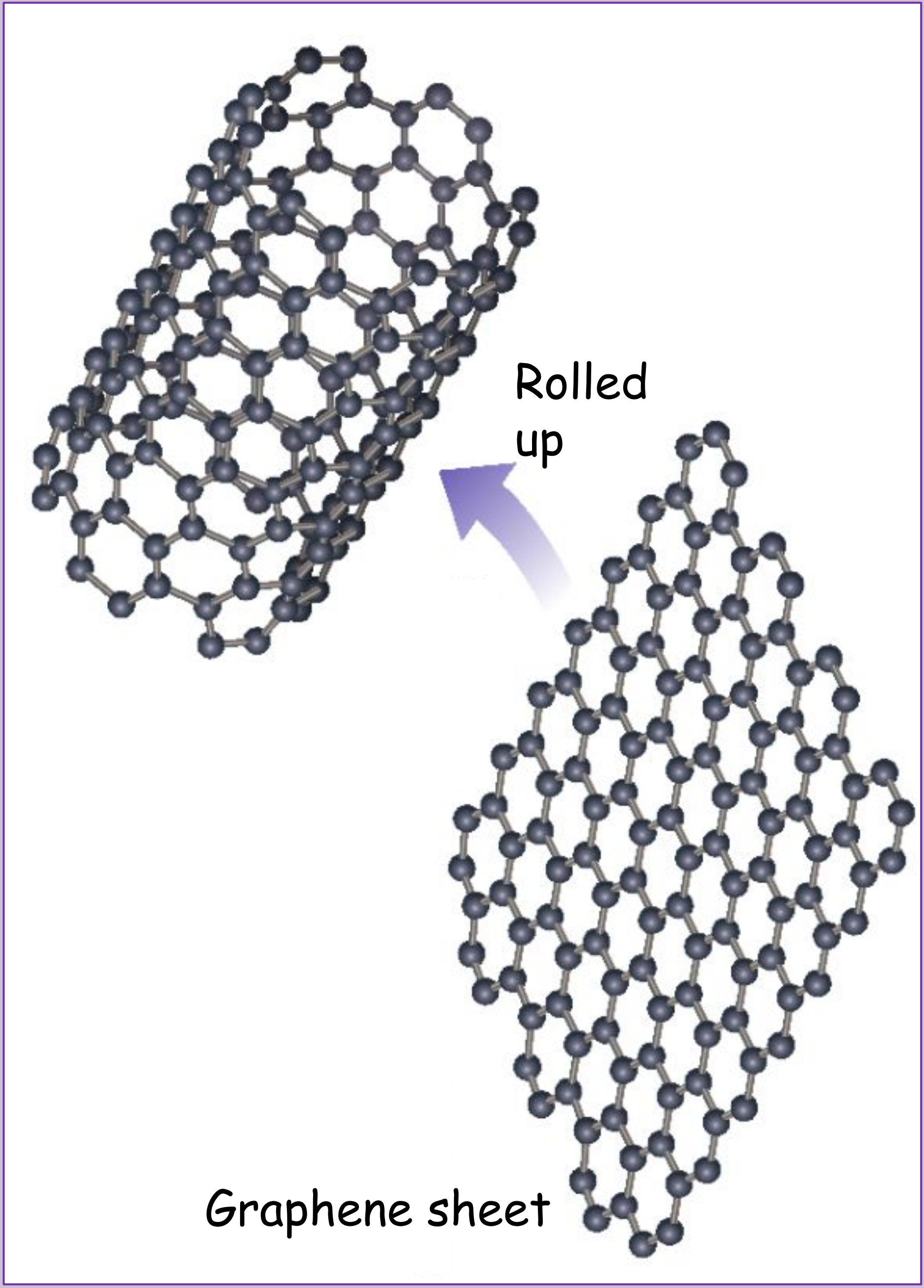
GRAPHENE

Similar to fullerenes but it is **not a simple molecule**.

Consists of a sheet of carbon atoms with no fixed formula. The sheet is **one atom thick** making it the **lightest** known material. Its **covalent** bonds make it extremely **strong** though.

It also allows free electrons to move across its surface and so is a **good electrical conductor**.

Graphene can be a **sheet** or rolled up into a **tube**.

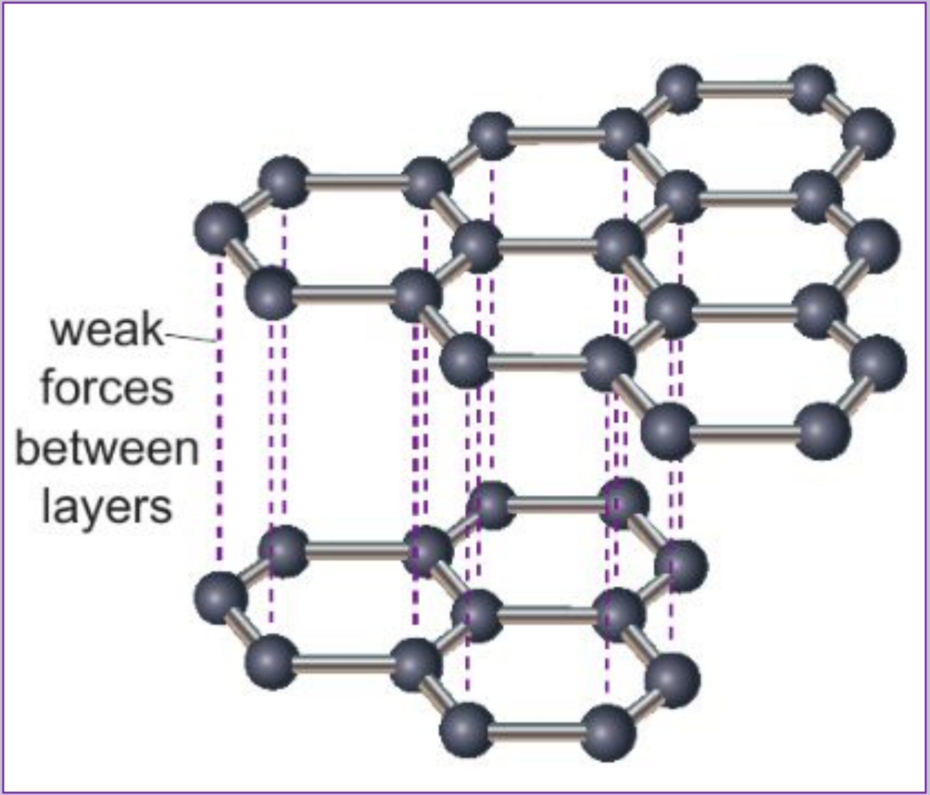


GRAPHITE

Not a simple molecule but a **covalent, giant molecular structure**. Made up of huge three-dimensional networks of carbon atoms linked by **covalent** bonds.

High melting point due to the strong covalent bonds.

Layered structure that means not all electrons are held in covalent bonds and are free to move, allowing it to **carry an electrical current**. It is often used in electrodes



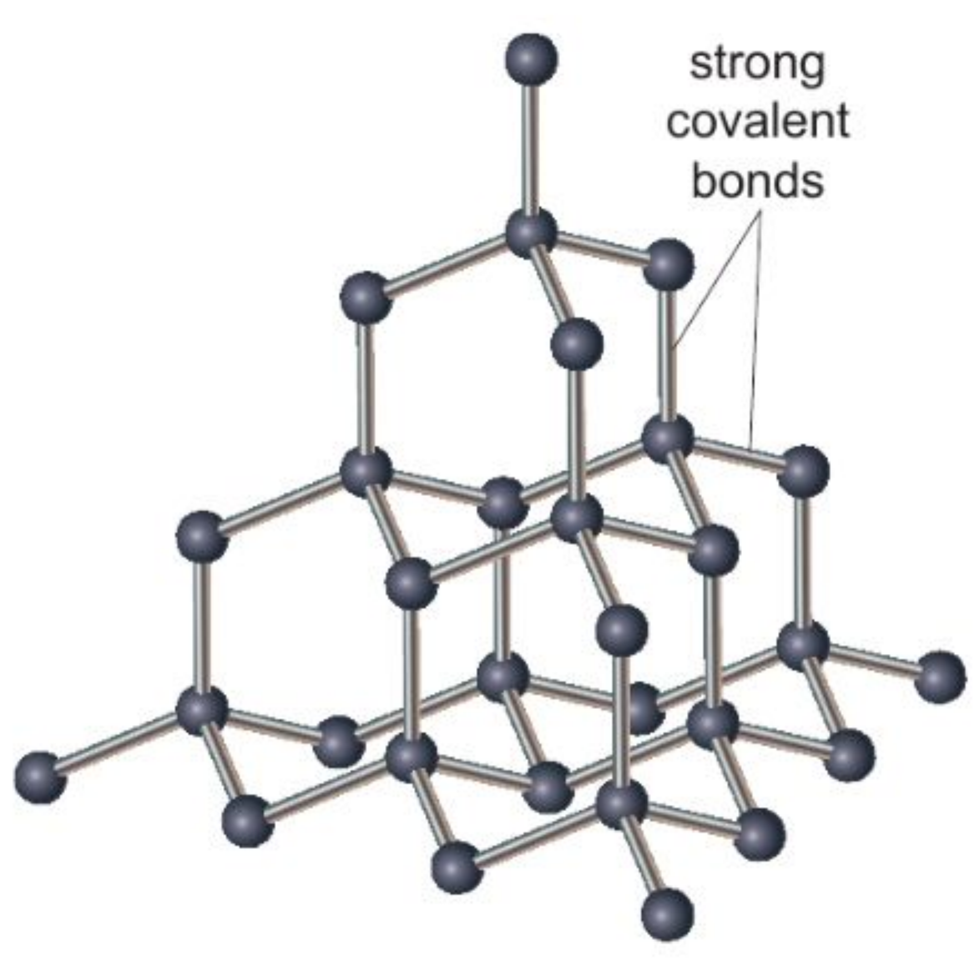
Weak forces of attraction hold the sheets together which allows the layers to **slide** past each other, making it quite **soft** and useful as a lubricant.

DIAMOND

Not a simple molecule but a **covalent, giant molecular structure**. Made up of huge three-dimensional networks of carbon atoms linked by **covalent** bonds.

High melting point due to the strong covalent bonds.

Network of carbon atoms in a **tetrahedral** arrangement, joined by strong **covalent** bonds, making diamond **very hard**. This makes it useful for tools to cut things.



No free charged electrons so is a good electrical **insulator**.

Answer the questions correctly to reveal an allotrope of carbon.

How soon can you work out which it is?

What type of bonding occurs between carbon atoms in diamond?	Why is graphite quite soft?	Which allotropes of carbon can conduct electricity?
How many carbon atoms are in a 'Bucky ball'?	What is the arrangement of carbon atoms in diamond?	How many atoms thick is a sheet of graphene?
Why do graphite and diamond have high melting points?	What are there between the layers in graphite?	What shapes do fullerenes often take?